POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessors

Semester: Spring
Year : 2014
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1) What are the essential differences between: 8
   i. Von Neumann and Harvard Architecture.
   ii. Microprocessor and Microcontroller.
2) Define flags and addressing modes. Explain the role of all the flag bits in 8085 microprocessor with essential examples. 7
3) Explain the function of the following 8085 program. What will be the output of the program displayed at port 01H when the program is executed if the value of BYTE1 is equal to FFH. 5

```
        MVI A,BYTE1
        ORA A
        JM   OUTPRT
        OUT  01H
        HLT
OUTPRT: CMA
        ADI  01H
        OUT  01H
        HLT
```

b) Specify the function of the following instructions from 8085 instruction set. 5
   i. LXI
   ii. LDAX
   iii. JC
   iv. PUSH
   v. SBB
c) Write a 8085 program to sort given 10 numbers from memory location 2200H in the ascending order. 5

3. a) What is the significance of Timing Diagram? Draw the labeled timing diagram for the instruction OUT 4FH. 8
   b) Explain how 1KB RAM, 2KB ROM and 4KB EPROM are interfaced with 8085 microprocessor and also find the address range for each chips. 7
4. a) Explain control word format of 8255 PPI for I/O mode. 7
   b) Draw the functional block diagram of 8086 Microprocessor and explain it and illustrate how its architecture is faster than its predecessors. 8
5. a) Write an 8086 ALP for MASM to find the square of a given number. 8
   b) Write a program to reverse the given string of 8086. 7
6. a) Define "Assembler Directives". Explain the following assembler directives along with their format: The Memory Model Definition, The PROC directive, The .STARTUP directive, The .EXIT directive and The END directive. 7
   b) Two computers are to be connected with each other via modems Using RS-232 standards, explain the connection. How can they be connected directly without any modems using RS 232 standards? 8
7. Write short notes on: (Any two) 2×5
   a) One pass and two pass assembler.
   b) RS 422A.
   c) Direct Memory Access.

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